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Applications of an energy-momentum tensor in non-linear elastostatics

P. CHADWICK
School of Mathematics and Physics, University of East Anglia, Norwich, England

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ABSTRACT

For equilibrium states of elastic materials some general formulae of conservation type have been established in recent papers by Knowles and Sternberg and by Green. It is shown that these results arise naturally from the application of standard integral identities to an energy-momentum tensor first introduced into elastostatics by Eshelby. A duality is exhibited between the energy-momentum tensor and the Cauchy stress which leads directly to inverse deformation relations for elastic solids due originally to Shield.

RÉSUMÉ

Knowles et Sternberg et Green ont récemment établi des formules générales du genre conservation pour des états d'équilibre dans des matières élastiques. On démontre que ces résultats découlent naturellement de l'application des identités intégrales standard à un tenseur d'énergie-moment qui a été introduit dans l'élastostatique pour la première fois par Eshelby. On montre qu'il existe une dualité entre le tenseur d'energie-moment et le tenseur de contraintes de Cauchy qui mène directement à des relations inverses de déformation pour des solides élastiques que Shield fut le premier à obtenir.

1. Introduction

The advantages to be gained by introducing an energy-momentum tensor into the theory of defects (such as voids, cracks and inhomogeneities) in elastic bodies have been demonstrated by Eshelby in a series of papers [1], [2], [3]. Mention of Eshelby's work is to be found in a recent study by Knowles and Sternberg [4] of formulae of conservation type in elastostatics, but the relevance of the concept of an energy-momentum tensor to theoretical questions concerning finite deformations seems not to have been fully appreciated. This is perhaps surprising in view of the attention specifically devoted by Eshelby to non-linear elastic behaviour in the latest of the above-mentioned articles [3; see also 2, pp. 103–104].

In the present paper we focus attention on a tensor $\boldsymbol{\Sigma}$ which, in respect of elastic materials possessing a strain-energy function, is a modification of Eshelby's energy-momentum tensor $\boldsymbol{P}$ nevertheless enjoying the same physical interpretation as $\boldsymbol{P}$ in terms

of the force (or energy release rate) accompanying the translation of a defect. The definition and basic properties of $\boldsymbol{\Sigma}$ are set out in section 3 and it is then proved, in section 4, that the two conservation formulae for finite deformations of elastic materials established by Knowles and Sternberg [4, section 4], together with a third result of similar type given by Eshelby [3, section 4], and independently by Green [5, section 2], are simple consequences of standard integral identities applied to $\boldsymbol{\Sigma}$. The directness of this approach is in marked contrast to the elaborate theory which Knowles and Sternberg developed in seeking an appropriate mathematical setting for their results.

Included in Eshelby's work [3, section 7] are indications of a reciprocal relationship between an energy-momentum tensor and the Cauchy stress $\boldsymbol{\sigma}$. It is shown in section 5 that there is in fact a precise duality between $\boldsymbol{\Sigma}$ and $\boldsymbol{\sigma}$, each of these tensors being transformed into the other when the reference and deformed configurations of the material are interchanged. Associated with this property are dual constitutive equations which are derived for both unconstrained and incompressible elastic materials. Substitution of these equations into the referential and spatial forms of the equilibrium equation is shown finally to reproduce inverse deformation results originally discovered by Shield [6].

2. Integral formulae for a solenoidal tensor field

We begin by reviewing some well-known results which are basic to the derivation of the conservation-type formulae given in section 4.

Let D be a domain of a three-dimensional Euclidean point space $\mathfrak{E}$ and let $\mathscr{R}$ be the set of all regular regions contained in D. A typical member of $\mathscr{R}$ is denoted by R, its boundary by ∂R, and the outward unit vector normal to ∂R by $\boldsymbol{N}$. The position of a representative point of $\mathfrak{E}$ relative to an arbitrary origin O is specified by the vector $\boldsymbol{X}$ and we use the notations Grad and Div for the gradient and divergence operators with reference to the position $\boldsymbol{X}$.

Suppose now that $\boldsymbol{T}$ is a tensor field which is continuously differentiable in D and let $\boldsymbol{t}$ be the vector field on ∂R defined by $\boldsymbol{t} = \boldsymbol{T}^T\boldsymbol{N}$, the superscript T signifying transposition. The results which we require state that if $\operatorname{Div}\boldsymbol{T} = \boldsymbol{0}$ in D then

$$\int_{\partial R} \boldsymbol{t}\,dA = \boldsymbol{0} \qquad \forall R \in \mathscr{R} \tag{1}$$

and

$$\int_{\partial R} \boldsymbol{X} \otimes \boldsymbol{t}\,dA = \int_R \boldsymbol{T}\,dV \qquad \forall R \in \mathscr{R}; \tag{2}$$

furthermore, if $\boldsymbol{T}$ is symmetric,

$$\int_{\partial R} \boldsymbol{X} \wedge \boldsymbol{t}\,dA = \boldsymbol{0} \qquad \forall R \in \mathscr{R}. \tag{3}$$

Here V and A denote volume in R and area on ∂R respectively, while $\otimes$ indicates a tensor product and $\wedge$ a vector product of vectors.

Equations (1) to (3) are consequences of the divergence theorem which we quote in the tensorial form [1]

$$\int_{\partial R} \boldsymbol{u} \otimes (\boldsymbol{T}^T \boldsymbol{N}) dA = \int_R \{\boldsymbol{u} \otimes \text{Div } \boldsymbol{T} + (\text{Grad } \boldsymbol{u})\boldsymbol{T}\} dV. \tag{4}$$

In (4) the vector field $\boldsymbol{u}$ and the tensor field $\boldsymbol{T}$ are required to be continuous in R and continuously differentiable in $R - \partial R$. Thus, if $\boldsymbol{u}$ and $\boldsymbol{T}$ are continuously differentiable and Div $\boldsymbol{T} = \boldsymbol{0}$ in D we deduce that

$$\int_{\partial R} \boldsymbol{u} \otimes \boldsymbol{t} dA = \int_R (\text{Grad } \boldsymbol{u})\boldsymbol{T} dV \qquad \forall R \in \mathscr{R}.$$

On choosing $\boldsymbol{u}$ to be an arbitrary constant vector we obtain formula (1), and the choice $\boldsymbol{u} = \boldsymbol{X}$ leads directly to (2). When $\boldsymbol{T}$ is symmetric the result of forming the skew-symmetric part of each side of equation (2) is

$$\int_{\partial R} (\boldsymbol{X} \otimes \boldsymbol{t} - \boldsymbol{t} \otimes \boldsymbol{X}) dA = \boldsymbol{O} \qquad \forall R \in \mathscr{R},$$

and since the axial vector of the integrand is $-\boldsymbol{X} \wedge \boldsymbol{t}$ [7, p. 30], formula (3) follows.

3. An energy-momentum tensor for elastic materials

We consider a material body $\mathscr{B}$ with reference configuration B and deformed configuration b and designate by $\boldsymbol{X}$ and $\boldsymbol{x}$ respectively the referential and spatial positions of a typical particle relative to a common origin O. The deformation gradient tensor $\boldsymbol{F}$ and the right Cauchy-Green strain tensor $\boldsymbol{C}$ associated with the deformation $B \to b$, assumed smooth, are given by

$$\boldsymbol{F} = \text{Grad } \boldsymbol{x}, \qquad \boldsymbol{C} = \boldsymbol{F}^T \boldsymbol{F}, \tag{5}$$

the notations for the gradient and divergence operators introduced in section 2 now being understood to relate to the referential position $\boldsymbol{X}$. The Cauchy, nominal and second (symmetric) Piola-Kirchhoff stress tensors are denoted in turn by $\boldsymbol{\sigma}$, $\boldsymbol{s}$ and $\boldsymbol{\tau}$, and we remind the reader of the connexions

$$\boldsymbol{\sigma} = J^{-1} \boldsymbol{F} \boldsymbol{s} = J^{-1} \boldsymbol{F} \boldsymbol{\tau} \boldsymbol{F}^T, \tag{6}$$

with $J = \det \boldsymbol{F}$ (> 0).

It is assumed henceforth that $\mathscr{B}$ is composed of a homogeneous elastic material with strain-energy function $W(\boldsymbol{F})$, measured per unit volume in the reference configuration B. We shall be concerned only with situations in which the deformation $B \to b$ is performed isothermally, wherefore W can be identified with the Helmholtz free energy of the ma-

[1] Equation (4) may be derived by applying the standard form of the divergence theorem to the vector field $(\boldsymbol{u} \cdot \boldsymbol{a})\boldsymbol{T}\boldsymbol{b}$, where $\boldsymbol{a}$ and $\boldsymbol{b}$ are arbitrary (constant) vectors [7, pp. 43–44].

terial evaluated at the fixed temperature in question. For an unconstrained material the constitutive equation is conveniently written as

$$\boldsymbol{s} = (\partial_{\boldsymbol{F}} W)^T, \tag{7}$$

where $\partial_{\boldsymbol{F}} W$ stands for the tensor whose components relative to a particular basis are the partial derivatives of W with respect to the corresponding components of $\boldsymbol{F}$. In the case of an incompressible material equation (7) must be replaced by

$$\boldsymbol{s} = -p\boldsymbol{F}^{-1} + (\partial_{\boldsymbol{F}} W)^T, \qquad \text{with } J = 1, \tag{8}$$

p being a scalar, interpretable as a pressure, which is not determined *a priori* by the deformation.

In writing equations in component form we select a fixed orthonormal basis in relation to which the referential and spatial coordinates are X_α and x_i respectively. Thus the component forms of equations $(5)_1$, (7) and (8) are

$$F_{i\alpha} = \partial x_i/\partial X_\alpha, \qquad s_{\alpha i} = \partial W/\partial F_{i\alpha}, \qquad s_{\alpha i} = -pF_{\alpha i}^{-1} + (\partial W/\partial F_{i\alpha}). \tag{9}$$

Greek and Latin subscripts have the range 1, 2, 3 and the summation convention applies to repeated indices.

The discussion which follows deals exclusively with elastostatics. Thus $\mathscr{B}$ is taken to be in equilibrium in the deformed state b and we suppose, moreover, that it can be so maintained without the aid of body forces. The equilibrium equation, in the referential description, is therefore

$$\text{Div } \boldsymbol{s} = \boldsymbol{0}, \tag{10}$$

and in component form this reads

$$\partial s_{\pi i}/\partial X_\pi = 0. \tag{11}$$

The energy-momentum tensor $\boldsymbol{\Sigma}$ on which the subsequent developments hinge is defined by

$$\boldsymbol{\Sigma} = W\boldsymbol{I} - \boldsymbol{s}\boldsymbol{F} = W\boldsymbol{I} - \tau\boldsymbol{C}, \tag{12}$$

where $\boldsymbol{I}$ is the identity tensor. An easy calculation shows that $\boldsymbol{\Sigma}$ is solenoidal. For an unconstrained elastic material we proceed, in terms of components, as follows:

$$\frac{\partial \Sigma_{\pi\alpha}}{\partial X_\pi} = \frac{\partial W}{\partial X_\alpha} - \frac{\partial}{\partial X_\pi}(s_{\pi p} F_{p\alpha}) = \frac{\partial W}{\partial F_{p\pi}} \frac{\partial F_{p\pi}}{\partial X_\alpha} - s_{\pi p} \frac{\partial F_{p\alpha}}{\partial X_\pi}$$

$$= s_{\pi p}\left(\frac{\partial x_p}{\partial X_\alpha \partial X_\pi} - \frac{\partial^2 x_p}{\partial X_\pi \partial X_\alpha}\right) = 0,$$

successive use being made of equations $(12)_1$, (11), $(9)_2$ and $(9)_1$. When the incompressibility constraint applies the substitution for $\partial W/\partial F_{p\pi}$ comes from equation $(9)_3$ and, since $J \equiv 1$, we find that

$$\frac{\partial \Sigma_{\pi\alpha}}{\partial X_\pi} = p \frac{\partial X_\pi}{\partial x_p} \frac{\partial F_{p\pi}}{\partial X_\alpha} = p \frac{\partial}{\partial x_p}(J^{-1} F_{p\alpha}) = 0$$

(see, for example, [7, p. 59]). Thus, in both cases[2],

$$\text{Div } \boldsymbol{\Sigma} = \mathbf{0}. \tag{13}$$

If the elastic material under consideration is isotropic relative to the reference configuration B the symmetric stress and strain tensors $\boldsymbol{\tau}$ and $\boldsymbol{C}$ have a common set of principal axes at each particle in all deformations from B. Hence $\boldsymbol{\tau C} = \boldsymbol{C\tau}$ (see, for example, [8, p. 265]) and it follows from $(12)_2$ that $\boldsymbol{\Sigma}$ is symmetric. But when the material is anisotropic in relation to B, $\boldsymbol{\Sigma}$ is generally non-symmetric.

The tensor $\boldsymbol{\Sigma}$ defined by equations (12) is closely related to the energy-momentum tensor $\boldsymbol{P}$ first introduced into elasticity theory, in the context of infinitesimal deformations, by Eshelby [1, sections 7 to 9] and subsequently given a central place in the same author's development of the continuum approach to the study of defects in solid media [2], [3]. In reference [3] Eshelby has given a detailed discussion of the energy-momentum tensor within the exact framework of non-linear elastostatics and the expression for $\boldsymbol{P}$ adopted there is, in the present notation,

$$\boldsymbol{P} = W\boldsymbol{I} - \boldsymbol{F}^T\boldsymbol{s}^T + \boldsymbol{s}^T = \boldsymbol{\Sigma}^T + \boldsymbol{s}^T. \tag{14}$$

It follows from $(14)_2$ that $\boldsymbol{P}$ is generally non-symmetric, even for isotropic elastic materials, and this partly explains our preference here for the tensor $\boldsymbol{\Sigma}$. We note, however, that a variant of $\boldsymbol{P}$, denoted by $\boldsymbol{P}^*$ and adopted by Eshelby [3, section 5] in situations where body forces are operative, is simply the transpose[3] of $\boldsymbol{\Sigma}$. All the properties of $\boldsymbol{\Sigma}$ given in this paper are accordingly implicit in sections 3 to 5 of [3].

4. Formulae of conservation type

Since the tensor field $\boldsymbol{\Sigma}$ is continuously differentiable and solenoidal in the interior of the reference configuration B it now follows from the integral identities (1) and (2) that

$$\int_{\partial R} \boldsymbol{\Sigma}^T \boldsymbol{N} dA = \mathbf{0} \tag{15}$$

and

$$\int_{\partial R} \boldsymbol{X} \otimes (\boldsymbol{\Sigma}^T \boldsymbol{N}) dA = \int_R \boldsymbol{\Sigma} dV \tag{16}$$

for an arbitrary regular subregion R of B. Additionally, if the elastic material constituting the body is isotropic in relation to B we infer from the identity (3) that

[2] In the absence of body forces equation (13) holds only for homogeneous materials. This restriction does not apply of course to (10) (or to equation (29) below).

[3] The fact that it is $(\boldsymbol{\Sigma} + \boldsymbol{s})^T$ and $\boldsymbol{\Sigma}^T$, not $\boldsymbol{\Sigma} + \boldsymbol{s}$ and $\boldsymbol{\Sigma}$, that are equivalent to $\boldsymbol{P}$ and $\boldsymbol{P}^*$ merely reflects different conventions in the definition of stress. With regard to the Cauchy stress, for example, σ_{ij} is taken in [3] and [4] to be the i-component of the force per unit area in the deformed configuration b acting on a surface segment with outward unit normal in the j-direction. In the present work, and in [5], this statement holds with i and j interchanged.

$$\int_{\partial R} \boldsymbol{X} \wedge (\boldsymbol{\Sigma}^T \boldsymbol{N}) dA = \boldsymbol{0}. \tag{17}$$

Equations (15) and (17) lead directly to the two conservation formulae for finite deformations of elastic materials which have been proved by Knowles and Sternberg [4, section 4] and re-derived, on the basis of simpler arguments, by Green [5, section 2]. Let $\boldsymbol{p} = \boldsymbol{s}^T \boldsymbol{N}$ be the stress vector on ∂R associated with the nominal stress $\boldsymbol{s}$. Then, on entering into (15) and (17) the definition $(12)_1$, we find that

$$\int_{\partial R} (\boldsymbol{W}\boldsymbol{N} - \boldsymbol{F}^T \boldsymbol{p}) dA = \boldsymbol{0} \tag{18}$$

and

$$\int_{\partial R} \boldsymbol{X} \wedge (\boldsymbol{W}\boldsymbol{N} - \boldsymbol{F}^T \boldsymbol{p}) dA = \boldsymbol{0}. \tag{19}$$

Apart from differences of notation, these results agree with Green's equations (2.8) and (2.9), while the conservation formulae (4.34) and (4.35) of Knowles and Sternberg are recovered on adding to the left sides of (18) and (19) the terms

$$\int_{\partial R} \boldsymbol{p} dA \quad \text{and} \quad \int_{\partial R} \boldsymbol{x} \wedge \boldsymbol{p} dA. \tag{20}$$

The first of the integrals (20) is zero on account of equations (1) and (10), and in order to confirm that the second integral is likewise null we return to the divergence theorem (4) and set $\boldsymbol{u} = \boldsymbol{x}$, $\boldsymbol{T} = \boldsymbol{s}$. In view of equation (10) and the definitions $(5)_1$ and $(6)_1$ we obtain

$$\int_{\partial R} \boldsymbol{x} \otimes \boldsymbol{p} dA = \int_R J \boldsymbol{\sigma} dV,$$

and the desired conclusion is reached by forming the skew-symmetric part of each side. We point out once more that formula (18) applies to homogeneous elastic materials in general, whereas (19) holds only for isotropic materials.

Eshelby [3, section 4] has shown that the energy-momentum tensor $\boldsymbol{P}$ defined by equation $(14)_1$ has the following direct physical interpretation. Suppose that $\mathscr{B}$ contains a defect d which, in the reference configuration B, is surrounded and separated from any other defects by the surface ∂R. Then the vector

$$\boldsymbol{\Xi} = \int_{\partial R} \boldsymbol{P}\boldsymbol{N} dA$$

represents the force on d in the sense that a small displacement $\delta\boldsymbol{\xi}$ of this defect gives rise to a change of amount $-\boldsymbol{\Xi} \cdot \delta\boldsymbol{\xi}$ in the total energy (i.e. the strain-energy content of R plus the potential energy of the tractions acting on ∂R). Equivalently, $\boldsymbol{\Xi}$ specifies the rate of energy release when d is translated relative to the body (cf. [9, pp. 205–213]). In view of the relation $(14)_2$ the difference between the integrals over ∂R of $\boldsymbol{P}\boldsymbol{N}$ and $\boldsymbol{\Sigma}^T \boldsymbol{N}$ is given by $(20)_1$ which has just been shown to be zero. The modified energy-momentum tensor $\boldsymbol{\Sigma}$ can therefore be interpreted in the same way as $\boldsymbol{P}$ (cf. [3, section 5]). If the region R contains no defect, then $\boldsymbol{\Xi} = \boldsymbol{0}$ and we recover equation (15). The first of the conserva-

tion laws established by Knowles and Sternberg is thus seen to be implicit in the earlier work of Eshelby.

Equation (16) is more general in form than results previously published, but we can deduce from it a known formula of conservation type by forming the trace of each side. The definition $(12)_1$, with $(9)_1$ and (11), yields

$$\operatorname{tr} \boldsymbol{\Sigma} = 3W - s_{\pi p} \frac{\partial x_p}{\partial X_\pi} = 3W - \frac{\partial}{\partial X_\pi} (s_{\pi p} x_p) = 3W - \operatorname{Div} (\boldsymbol{s}\boldsymbol{x}),$$

and, after applying to the volume integral of $\operatorname{Div}(\boldsymbol{s}\boldsymbol{x})$ the standard form of the divergence theorem, we find that

$$3 \int_R W dV = \int_{\partial R} (\boldsymbol{\Sigma}\boldsymbol{X} + \boldsymbol{s}\boldsymbol{x}) \cdot \boldsymbol{N} dA \tag{21}$$

$$= \int_{\partial R} \{W\boldsymbol{X} \cdot \boldsymbol{N} + \boldsymbol{p} \cdot (\boldsymbol{x} - \boldsymbol{F}\boldsymbol{X})\} dA. \tag{22}$$

Equation (21), with the integrand on the right expressed in terms of $\boldsymbol{P}$, has been obtained by Eshelby [3, equation (16)] and equation (22), independently, by Green [5, equation (2.10)]; they provide alternative expressions for the strain-energy content of the region R in terms of stress and deformation data on the boundary ∂R. Some applications of (22) to particular deformations of unconstrained elastic bodies are described in sections 3 to 6 of Green's paper. The specialization of equation (22) appropriate to infinitesimal deformations has been derived by Knowles and Sternberg [4, equation (3.19)] in association with forms of the conservation formulae (18) and (19) valid in the same context [4]. Interpretations of all three formulae in terms of the energy-release rates accompanying the translation, rotation and expansion of voids and cracks have recently been supplied by Budiansky and Rice [11].

5. Duality of the energy-momentum tensor $\boldsymbol{\Sigma}$ and the Cauchy stress $\boldsymbol{\sigma}$

The fact that the modified energy-momentum tensor $\boldsymbol{\Sigma}$ has the physical dimensions of stress, allied to the results established in section 3, suggests that $\boldsymbol{\Sigma}$ has properties in relation to the reference configuration B analogous to those borne by the Cauchy stress $\boldsymbol{\sigma}$ to the current configuration b. In fact a formal reciprocity between $\boldsymbol{\Sigma}$ and $\boldsymbol{\sigma}$ is immediately apparent from the definition $(12)_1$. Let $\boldsymbol{f}$ denote the inverse of the deformation gradient (otherwise $\boldsymbol{F}^{-1}$), and let $w(\boldsymbol{f})$ be the strain energy per unit volume in b, defined by

$$w(\boldsymbol{f}) = jW(\boldsymbol{f}^{-1}); \qquad W(\boldsymbol{F}) = Jw(\boldsymbol{F}^{-1}), \tag{23}$$

where $j = \det \boldsymbol{f} = J^{-1}$. Then, by virtue of equation $(6)_1$, there follow from $(12)_1$ the dual relations

$$\boldsymbol{\Sigma} = W(\boldsymbol{F})\boldsymbol{I} - J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}; \qquad \boldsymbol{\sigma} = w(\boldsymbol{f})\boldsymbol{I} - j\boldsymbol{f}^{-1}\boldsymbol{\Sigma}\boldsymbol{f}, \tag{24}$$

[4] These results also appear in an earlier paper by Günther [10, section 2]. I am grateful to Professor Eshelby for this reference.

indicating that $\boldsymbol{\sigma}$ is transformed into $\boldsymbol{\Sigma}$, and $\boldsymbol{\Sigma}$ into $\boldsymbol{\sigma}$, when the configurations B and b are interchanged (cf. [12, equations (7), (8) and (5)]).

When the material is isotropic relative to the reference configuration B the Cauchy stress $\boldsymbol{\sigma}$ and the left Cauchy-Green strain tensor $\boldsymbol{B} = \boldsymbol{F}\boldsymbol{F}^T$ have a common set of principal axes in b. The duality between $\boldsymbol{\Sigma}$ and $\boldsymbol{\sigma}$ which has just been noticed indicates that in this case $\boldsymbol{\Sigma}$ and $\boldsymbol{f}\boldsymbol{f}^T$ (which, from equation $(5)_2$, is $\boldsymbol{C}^{-1}$) should have common principal axes in B. Since compositional commutativity of two symmetric tensors is a sufficient as well as a necessary condition for coaxiality [8. p. 265], this property is confirmed by the relation

$$\boldsymbol{C}^{-1}\boldsymbol{\Sigma}-\boldsymbol{\Sigma}\boldsymbol{C}^{-1} = J\boldsymbol{F}^{-1}\boldsymbol{B}^{-1}(\boldsymbol{B}\boldsymbol{\sigma}-\boldsymbol{\sigma}\boldsymbol{B})(\boldsymbol{F}^{-1})^T,$$

derived from equation $(24)_1$.

An alternative, and more direct, expression of the duality between $\boldsymbol{\Sigma}$ and $\boldsymbol{\sigma}$ is reached by making use of equations (7) and (8). For unconstrained materials dual constitutive equations are obtained in the form

$$\boldsymbol{\Sigma} = j^{-1}\boldsymbol{f}(\partial_{\boldsymbol{f}} w)^T; \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{F}(\partial_{\boldsymbol{F}} W)^T, \tag{25}$$

and the corresponding results for an incompressible material are

$$\boldsymbol{\Sigma} = -P\boldsymbol{I}+\boldsymbol{f}(\partial_{\boldsymbol{f}} w)^T; \qquad \boldsymbol{\sigma} = -p\boldsymbol{I}+\boldsymbol{F}(\partial_{\boldsymbol{F}} W)^T, \tag{26}$$

with

$$P = -p-w(\boldsymbol{f}). \tag{27}$$

Equation $(25)_2$ follows directly from $(6)_1$ and (7). In order to verify equation $(25)_1$ we first substitute $(25)_2$ into the relation $(24)_1$ and call upon the well-known differentiation formula $\partial_{\boldsymbol{F}} J = J\boldsymbol{f}^T$. The resulting expression for $\boldsymbol{\Sigma}$ can be cast into the form

$$\boldsymbol{\Sigma} = -J\{\partial_{\boldsymbol{F}}(J^{-1}W)\}^T\boldsymbol{F}$$

and, in view of the definition $(23)_1$, this becomes

$$\boldsymbol{\Sigma} = -j^{-1}(\partial_{\boldsymbol{F}} w)^T\boldsymbol{F}.$$

The final step consists of proving that

$$(\partial_{\boldsymbol{F}} w)^T\boldsymbol{F} = -\boldsymbol{f}(\partial_{\boldsymbol{f}} w)^T \tag{28}$$

and this is readily accomplished, in terms of components, with the aid of the formula

$$\partial f_{\alpha i}/\partial F_{j\beta} = -f_{\alpha j} f_{\beta i}$$

(see [6, equation $(2.4)_2$]).

When the elastic material under consideration is incompressible equation $(6)_1$ reduces to $\boldsymbol{\sigma} = \boldsymbol{F}\boldsymbol{s}$ which, combined with (8), yields equation $(26)_2$. Equations (23) and $(24)_1$ give, in this case,

$$W(\boldsymbol{F}) = w(\boldsymbol{f}) \qquad \text{and} \qquad \boldsymbol{\Sigma} = W(\boldsymbol{F})\boldsymbol{I}-\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F},$$

whence, with the use of $(26)_2$,

$$\boldsymbol{\Sigma} = \{p + w(\boldsymbol{f})\}\boldsymbol{I} - (\partial_{\boldsymbol{F}} w)^T \boldsymbol{F}.$$

Equation $(26)_1$, with (27), now follows on appealing once more to (28).

A general study of relationships in non-linear elastostatics which are dual to one another under interchange of the referential and spatial descriptions has been made by Shield [6] [5] and the main results obtained in section 3 of his paper can easily be recovered from equations (25) and (26). The dual of equation (13) is the spatial equilibrium equation

$$\operatorname{div} \boldsymbol{\sigma} = \mathbf{0}, \tag{29}$$

where the notations grad and div are now used for the gradient and divergence operators relative to the spatial position $\boldsymbol{x}$. On substituting from the dual constitutive equations (25) into (13) and (29) and employing standard manipulations we obtain

$$\operatorname{div} (\partial_{\boldsymbol{f}} w)^T = \mathbf{0}; \qquad \operatorname{Div} (\partial_{\boldsymbol{F}} W)^T = \mathbf{0}, \tag{30}$$

while, with the use of equations (26), the corresponding results for an incompressible material are found to be

$$-\operatorname{grad} P + \boldsymbol{f}^T \operatorname{div} (\partial_{\boldsymbol{f}} w)^T = 0; \qquad -\operatorname{Grad} p + \boldsymbol{F}^T \operatorname{Div} (\partial_{\boldsymbol{F}} W)^T = 0. \tag{31}$$

Equations (30) reproduce, in direct notation, equations (3.6) and (3.1) of Shield's paper, and (31) are equivalent to Shield's equations (3.11) and (3.10).

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